

# An Archetypal Analysis of ICML (2019) Conference

732A76 Research Project

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## Abstract

Archetypal Analysis (AA) was introduced by Cutler and Breiman in 1994 [1]. It is a method designed to extract distinct structural patterns, called archetypes, from observational data, where each data point is approximated as a convex mixture of these archetypes [2]. Although AA has been successfully applied in several domains, its potential as a topic modeling framework for scientific literature remains relatively underexplored. In this project, we investigate the use of Archetypal Analysis as a topic modeling approach for analyzing research abstracts related to ICML conference-2019 7.3.1. thereby creating a self-referential analysis of the literature.

The experimental results demonstrate that Archetypal Analysis (AA) produces highly interpretable and sparse topic representations that effectively capture extremal semantic structures within the literature, often more distinctly than probabilistic or centroid-based methods. This work highlights the suitability of Archetypal Analysis as a geometric framework for topic modeling and provides insights into the structure and evolution of research trends within the AA research community. The implementation and experimental results are publicly available at [https://github.com/AshenWELI/Archetypal\\_Analysis](https://github.com/AshenWELI/Archetypal_Analysis).

## 1 Introduction

Archetypal Analysis (AA) [1], introduced by Cutler and Breiman [1], provides the theoretical foundation for this project. According to their work, the original motivation for Archetypal Analysis arose from a practical problem in 1988, when the Swiss Army needed to design improved protective face masks for soldiers. This task was assigned to Flury and Riedwyl, who analyzed a large dataset of human head measurements and observed interesting geometric patterns in the data.[1]

A natural question that emerged from this study was whether a small number of “pure” head types, called *archetypes*, could explain the observed variability in the dataset. In other words, could the roughly 200 measured head shapes be represented as convex mixtures of a few extreme, idealized head shapes? To address this problem, Cutler and Breiman developed Archetypal Analysis.

In this work, we build on the theoretical framework of Archetypal Analysis and explore its application as a topic modeling method for scientific text data.

### 1.1 Research Aim and Targets

The primary aim of this project is to investigate Archetypal Analysis (AA) as a topic modeling framework for textual data, with a particular focus on scientific abstracts from the ICML 2019 conference. By applying AA to research literature related to Archetypal Analysis itself, this study seeks to uncover the underlying semantic structure and thematic organization of the corpus.

Specifically, this project aims to:

- Evaluate the suitability of Archetypal Analysis for modeling high-dimensional categorical text data.
- Reproduce and validate previous findings on text-based Archetypal Analysis reported in [3].

- Compare the performance and interpretability of AA against established topic modeling methods, including Non-Negative Matrix Factorization (NMF) 7.2.1,  $k$ -means clustering 7.2.2 [4] [5], and Latent Dirichlet Allocation (LDA) [6].
- Analyze the geometric and sparsity characteristics of AA-derived topic representations.

## 2 Methodology

This section describes the methodological framework employed in this research, outlining the procedures used to design, validate, and apply Archetypal Analysis (AA) for topic extraction from scientific abstracts. The methodology follows a structured analytical pipeline to ensure both technical rigor and empirical validity. As a prerequisite, we first provide an overview of Archetypal Analysis to establish the conceptual foundation of the proposed approach.

### 2.1 Archetypal Analysis

Let's consider a dataset represented by a matrix  $\mathbf{X} \in \mathbb{R}^{N \times M}$ , where  $N$  is the number of data samples and  $M$  is the number of features. The dataset can be written as

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\},$$

where each  $\mathbf{x}_i \in \mathbb{R}^M$  represents one observation.

The goal of Archetypal Analysis is to approximate the data matrix  $\mathbf{X}$  by a low-rank decomposition

$$\mathbf{X} \approx \mathbf{S}\mathbf{A},$$

where  $\mathbf{S} \in \mathbb{R}^{N \times K}$  and  $\mathbf{A} \in \mathbb{R}^{K \times M}$ .

Here,  $\mathbf{S}$  is the coefficient (or weight) matrix that expresses each data point as a convex combination of archetypes, and  $\mathbf{A}$  is the matrix containing the  $K$  archetypes themselves. [2] The common data representation in Archetypal Analysis expresses each data point  $\mathbf{x}_i$  as a convex combination of  $K$  archetypes  $a_1, a_2, \dots, a_K$ :

$$\mathbf{x}_i \approx \mathbf{A}^\top \mathbf{s}_i,$$

where  $s_i \geq 0$  and  $\sum_{j=1}^K s_{ij} = 1$ . The vector  $\mathbf{s}_i$  contains the weights of the convex combination, and these vectors are stacked row-wise to form the matrix  $\mathbf{S} \in \mathbb{R}^{N \times K}$ . The matrix  $\mathbf{A} \in \mathbb{R}^{K \times M}$  contains the archetypes  $a_1, a_2, \dots, a_K$  as its rows and is referred to as the design matrix of archetypes.

Each archetype can itself be expressed as a convex combination of the original data points:

$$\mathbf{a}_k = \mathbf{X}^\top \mathbf{c}_k,$$

where  $c_k \geq 0$  and  $\sum_{i=1}^N c_{ki} = 1$ . These coefficient vectors are stacked row-wise to form the matrix

$$\mathbf{C} \in \mathbb{R}^{K \times N}.$$

From these definitions, it follows that

$$\mathbf{A} = \mathbf{C}\mathbf{X}.$$

Due to the convexity constraints, both coefficient matrices  $\mathbf{S}$  and  $\mathbf{C}$  are row-stochastic, meaning that all entries are non-negative and each row sums to one.

### 2.1.1 Residual Sum of Squares (RSS)

The weight matrices are obtained by minimizing the Residual Sum of Squares (RSS) as the objective function, defined as

$$\text{RSS} = \|\mathbf{X} - \mathbf{S}\mathbf{C}\mathbf{X}\|_F^2 = \|\mathbf{X} - \mathbf{S}\mathbf{A}\|_F^2.$$

Here,  $\|\cdot\|_F^2$  denotes the squared Frobenius norm.

Optimizing the RSS in order to estimate the optimal weight matrices  $\mathbf{S}$  and  $\mathbf{C}$  leads to a non-convex optimization problem. However, the problem is convex in  $\mathbf{S}$  when  $\mathbf{C}$  is fixed, and convex in  $\mathbf{C}$  when  $\mathbf{S}$  is fixed [2].

For readers interested in the optimization procedure, the pseudocode is presented in Subsection 7.1.1. Further technical details can be found in the references [2], [3].

### 2.1.2 Geometric Interpretation

The minimization of the residual sum of squares (RSS) in Archetypal Analysis can be interpreted from a geometric perspective. In this view, the archetypes are selected as extremal points that span a large convex hull (illustrated as the light-blue region), while each data point is approximated by its projection onto this convex hull [1]. [2] The grey line segments represent the shortest Euclidean distances between each observation and its corresponding projection, and the colored points indicate the dominant archetype to which each observation is most strongly associated.

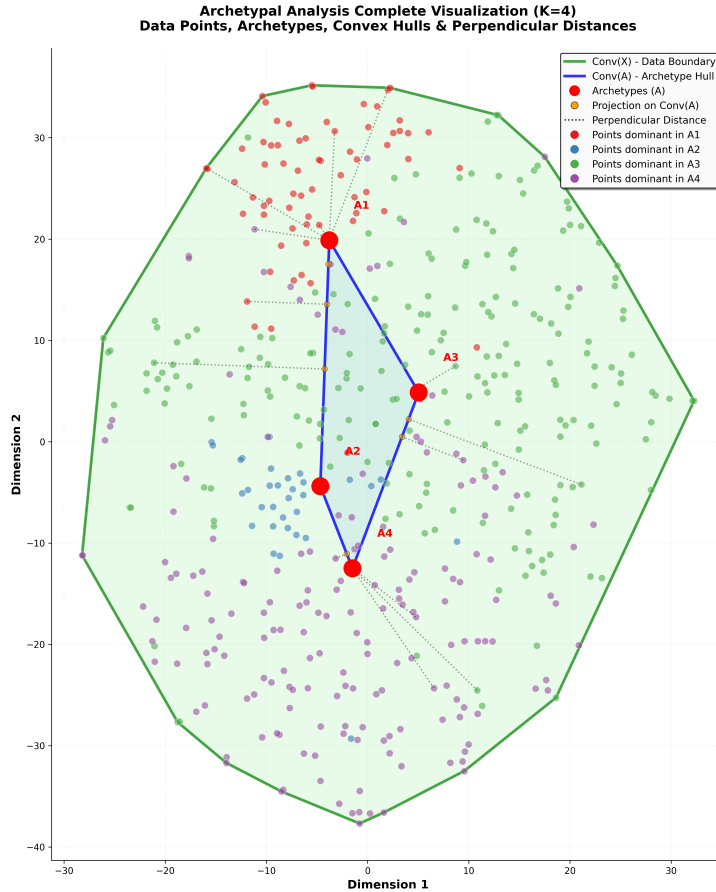


Figure 1: Visualization of Archetypal Analysis with  $K = 4$  on the conference abstract dataset. Colored points represent documents assigned to their dominant archetype, while Red big - Coloured Circles denote the learned archetypes. The figure illustrates a well-defined convex hull with clear cluster separation and a small number of deviating points.

Formally, this geometric interpretation expresses the RSS as the sum of squared distances between each data point and the convex hull of the archetypes [1], [3]:

$$\text{RSS} = \sum_{n=1}^N \min_{x' \in \text{conv}(\{a_1, \dots, a_K\})} \|x_n - x'\|_2^2,$$

where  $\text{conv}(\cdot)$  denotes the convex hull and  $a_1, \dots, a_K$  are the archetypes.

Beyond Euclidean distance, this geometric formulation naturally admits an angular interpretation. Since archetypes lie on the boundary of the convex hull and correspond to extremal directions in the data space, the cosine of the angle between a data point  $x_n$  and an archetype  $a_k$  provides a measure of directional alignment. A small cosine angle (i.e., high cosine similarity) indicates that the data point lies close to the ray defined by the archetype, even if its magnitude differs. Consequently, cosine similarity serves as a complementary interpretability measure that highlights semantic alignment with archetypal directions, particularly in high-dimensional text representations where vector norms vary substantially.

This directional viewpoint reinforces the interpretation of archetypes as extreme semantic patterns rather than centroids, distinguishing Archetypal Analysis from clustering methods such as  $k$ -means and aligning it more closely with geometric topic modeling approaches [1], [3].

**Experimental Workflow** The theoretical foundations required for the experiments are established in the preceding sub-sections. The first stage of this study involves implementing an Archetypal Analysis (AA) model from scratch. Owing to the non-convex nature of the AA optimization problem and its sensitivity to initialization, careful validation of the implementation is required. The algorithmic framework follows the formulation presented by Alcácer et al. [2], particularly Section II, with additional guidance drawn from the implementation details of Mørup and Hansen [3]. To ensure correctness and reproducibility, the custom AA model is validated using the NIPS Bag-of-Words dataset obtained from the UCI Machine Learning Repository, adopting an evaluation protocol consistent with that used in [3].

Once the AA implementation is validated, the second stage focuses on applying the model to the collected corpus of ICLR 2019 abstracts. The extracted topics (archetypes) are then compared against those produced by established topic modeling methods, including K-means clustering [4], [5], Non-Negative Matrix Factorization (NMF; see Section 7.2.1), and Latent Dirichlet Allocation (LDA)[6]. This comparative analysis allows us to assess differences in interpretability, sparsity, topic coherence, and reconstruction error across methods.

Together, these steps form a comprehensive methodology that ensures both the correctness of the AA implementation and the validity of its use as a topic modeling framework.

### 3 Experimental Results

This chapter presents the experimental evaluation of the proposed methodology. The experiments assess the effectiveness of Archetypal Analysis (AA) as a topic modeling approach for textual data and compare its performance with established baseline methods: Non-Negative Matrix Factorization (NMF) 7.2.1,  $k$ -means clustering [5] 7.2.2 [4], and Latent Dirichlet Allocation (LDA) [6]. The evaluation includes both validation experiments and large-scale experiments conducted on conference data.

#### 3.1 Data

Two publicly available text datasets are used in this study. The first dataset is employed for model validation and parameter calibration, while the second dataset consists of conference abstracts used for the main experimental analysis. Both datasets are of substantial size and enable reproducibility.

For validation experiments, the number of documents is restricted to a maximum of 500 to control computational cost while preserving sufficient semantic structure. The number of topics (archetypes) is fixed to  $K = 10$  across all models to ensure a fair comparison.

### 3.2 Model Validation with Test Data

Table 1 reports quantitative results obtained from the validation dataset. The evaluation metrics include reconstruction error, computational time, sparsity-related measures, and method-specific quality indicators.

Table 1: Summary statistics for topic modeling methods on validation data ( $K = 10$ , max\_docs=500).

| Metric                            | AA                    | NMF                   | $k$ -means            | LDA                |
|-----------------------------------|-----------------------|-----------------------|-----------------------|--------------------|
| Reconstruction Error (RSS)        | $3.31 \times 10^2$    | $3.17 \times 10^2$    | $3.47 \times 10^2$    | $2.27 \times 10^7$ |
| RSS per Document                  | $6.62 \times 10^{-1}$ | $6.33 \times 10^{-1}$ | $6.95 \times 10^{-1}$ | $4.53 \times 10^4$ |
| Computation Time (s)              | 2217.36               | 4.85                  | 2.42                  | 3.51               |
| Avg. Active Components / Document | 4.16                  | 6.21                  | 1.00                  | 1.53               |
| Silhouette Score ( $k$ -means)    | –                     | –                     | 0.021                 | –                  |
| Calinski–Harabasz Index           | –                     | –                     | 7.65                  | –                  |
| Perplexity (LDA)                  | –                     | –                     | –                     | 18863.92           |

#### 3.2.1 Geometric and Interpretability Analysis

Additional geometric and sparsity-related metrics are summarized in Table 2. These measures highlight structural differences in how each method represents documents in topic space.

Table 2: Comparative analysis of topic modeling methods on validation data.

| Metric                      | AA   | NMF  | $k$ -means | LDA   |
|-----------------------------|------|------|------------|-------|
| Non-zero entries in $S$ (%) | 43.5 | 66.5 | 10.0       | 100.0 |
| Avg. cosine angle (degrees) | 64.9 | 81.0 | 51.1       | 18.9  |

### 3.3 Experimental Results on Conference Data

This subsection presents results obtained from the main experimental run on the conference abstract dataset using  $K = 10$  topics. All methods are applied to the same normalized document-term matrix to ensure comparability.

Table 3 reports the primary quantitative metrics, capturing reconstruction quality, computational efficiency, and component usage.

Table 3: Summary statistics for topic modeling methods on conference data ( $K = 10$ ).

| Metric                            | AA                    | NMF                   | $k$ -means            | LDA                |
|-----------------------------------|-----------------------|-----------------------|-----------------------|--------------------|
| Reconstruction Error (RSS)        | $4.50 \times 10^2$    | $4.46 \times 10^2$    | $4.57 \times 10^2$    | $2.62 \times 10^6$ |
| RSS per Document                  | $8.99 \times 10^{-1}$ | $8.92 \times 10^{-1}$ | $9.15 \times 10^{-1}$ | $5.23 \times 10^3$ |
| Computation Time (s)              | 354.84                | 0.77                  | 0.86                  | 1.44               |
| Avg. Active Components / Document | 4.28                  | 4.79                  | 1.00                  | 9.83               |
| Silhouette Score ( $k$ -means)    | –                     | –                     | 0.008                 | –                  |
| Calinski–Harabasz Index           | –                     | –                     | 2.58                  | –                  |
| Perplexity (LDA)                  | –                     | –                     | –                     | 27328.50           |

### 3.3.1 Geometric and Interpretability Analysis

Table 4 summarizes key geometric and sparsity-related metrics for the conference dataset. These measures provide additional evidence supporting the main quantitative results reported earlier and offer insight into the interpretability of the extracted topic representations.

In particular, sparsity in the document–component matrix reflects how selectively a model represents documents using a small number of components, while cosine-angle separation captures the geometric distinctiveness between components. Together, these metrics provide a concrete geometric interpretation of model behavior on the conference abstracts.

Table 4: Geometric and sparsity comparison on conference data ( $K = 10$ ).

| Metric                      | AA   | NMF  | $k$ -means | LDA   |
|-----------------------------|------|------|------------|-------|
| Non-zero entries in $S$ (%) | 46.6 | 51.7 | 10.0       | 100.0 |
| Avg. cosine angle (degrees) | 73.8 | 79.1 | 64.1       | 46.4  |

## 4 Discussion

To maintain clarity and consistency in the analysis, the main results presented in this study are restricted to  $K = 10$  archetypes. Although experiments were conducted for multiple values of  $K$ , including larger settings, these additional results are reported in the Appendix and GitHub <sup>1</sup>. Since this work does not focus on a systematic study of model behavior as a function of  $K$ , comparisons across different values of  $K$  are not discussed in detail.

The first objective of the experiments was to validate the Archetypal Analysis (AA) model implemented from scratch. This validation was performed using the NIPS Bag-of-Words dataset, following the experimental setup described in [3]. The resulting word-distribution visualizations (Figure 4) show that AA and NMF produce qualitatively similar topic structures, with darker shades indicating higher word importance. This agreement provides an initial indication of the correctness of the custom AA implementation.

Reconstruction error, measured using the residual sum of squares (RSS), quantifies the amount of information lost during factorization. Lower RSS values indicate better reconstruction quality. Among the evaluated methods, NMF achieves the lowest RSS, while the proposed AA implementation attains the second-best performance, outperforming  $k$ -means and LDA. As expected, increasing the number of archetypes generally leads to a decrease in RSS across all methods. (Table 1)

A notable limitation of the AA model is its computational cost. The validation experiments report a substantially higher runtime for AA (2217.36 seconds) compared to the baseline methods. This is primarily due to the alternating constrained optimization steps and highlights an important direction for future work, such as algorithmic acceleration or parallelization.

Interpretability and sparsity are key strengths of Archetypal Analysis. The average number of active components per document for AA is 4.16, indicating that each document is explained by a small number of archetypes. This sparsity is further confirmed by the proportion of non-zero entries in the document–archetype matrix (43.5%), which closely matches the sparsity reported in [3] (42.2%). Such consistency further validates the implementation. (Table 3)

Geometric distinctiveness is assessed using the average cosine angle between components. While the cosine separation in the present implementation ( $64.9^\circ$ ) is slightly lower than that reported by Mørup and Hansen ( $80.31^\circ$ ), it still indicates well-separated semantic directions and meaningful archetypal structure.

Following the successful validation of the Archetypal Analysis (AA) model, the experiments were extended to a new dataset consisting of ICML 2019 conference abstracts obtained from OpenReview. The

<sup>1</sup>[https://github.com/AshenWELI/Archetypal\\_Analysis](https://github.com/AshenWELI/Archetypal_Analysis)

validation results confirmed that the AA model implemented from scratch operates correctly, allowing it to be reliably applied to unseen data.

The quantitative results for the ICML dataset are summarized in Table 2. For  $K = 10$ , the AA model achieves a reconstruction error (RSS) of  $4.5 \times 10^2$ , which is comparable to NMF and  $k$ -means and substantially lower than that of LDA, which reports the highest reconstruction error. When compared to the validation experiments on the NIPS dataset, the RSS values for the ICML corpus are noticeably higher across all methods.

This increase in RSS can be attributed primarily to dataset characteristics rather than model performance. In general, the RSS scales with the magnitude and dimensionality of the data, with  $E[\text{RSS}] \propto N \times M$ . Moreover, higher RSS values may arise when the dataset does not admit a low-rank approximation for a fixed  $K$ , for example due to overlapping topics or a long-tailed vocabulary distribution. Since both datasets were evaluated under identical experimental conditions ( $K = 10$  and a maximum of 500 documents), the elevated RSS observed for the conference abstracts is most plausibly explained by increased topic overlap. Scientific abstracts frequently span multiple themes, in contrast to the more homogeneous structure of the NIPS Bag-of-Words dataset.

In terms of computational efficiency, the AA model exhibits a notable reduction in runtime on the conference dataset compared to the validation experiments. Nevertheless, AA remains the most computationally expensive method among the evaluated approaches, with a runtime of approximately 354.84 seconds. In comparison, NMF completes in 0.77 seconds,  $k$ -means in 0.86 seconds, and LDA in 1.44 seconds. While this highlights a limitation of AA in terms of scalability, it is consistent with the iterative constrained optimization required by the method.

The average number of active components per document provides further insight into model behavior. This metric reflects how many topics or archetypes contribute meaningfully to each document. Lower values indicate sparser and more interpretable representations, while higher values suggest greater topic mixing. As expected,  $k$ -means yields the lowest value, since each document is assigned to exactly one cluster by design (see Section 7.2.2). Both AA and NMF activate a small number of components per document, indicating sparse and interpretable representations. This behavior confirms that meaningful low-dimensional structure exists within the conference dataset and further supports the suitability of Archetypal Analysis for modeling complex, multi-topic scientific texts. As discussed above, sparsity is a key indicator of topic interpretability and is clearly reflected in Table 4. Higher sparsity corresponds to cleaner topic assignments, whereas lower sparsity indicates increased topic mixing [2]. According to Table 4, Archetypal Analysis achieves a lower proportion of non-zero entries in the document–archetype coefficient matrix  $S$  compared to NMF (44.6% versus 51.7%). This demonstrates that AA yields a sparser document representation, with fewer active components per document. Such behavior is a direct consequence of the simplex constraints imposed on  $S$ , which encourage solutions near the faces of the simplex. In contrast, NMF permits denser mixtures due to the absence of a sum-to-one constraint on the document–topic coefficients.

The cosine-angle analysis further supports these findings. The average inter-component angles are  $73.8^\circ$ ,  $79.1^\circ$ ,  $64.1^\circ$ , and  $46.4^\circ$  for AA, NMF,  $k$ -means, and LDA, respectively. Angles exceeding  $60^\circ$  indicate well-separated semantic directions, whereas smaller angles correspond to greater overlap among components. From this perspective, both AA and NMF demonstrate strong topic separation, while LDA exhibits more moderate overlap (see Section 2.1). These results confirm that the proposed AA implementation performs competitively with established methods and effectively captures distinct semantic structures in the data.

These quantitative findings are further corroborated by the graphical visualizations presented in Figure 7 and additional figures in the Appendix. Together, the numerical metrics and visual evidence provide a consistent and interpretable picture of model behavior across both datasets.

Overall, Archetypal Analysis consistently produces sparse and interpretable document representations while maintaining competitive reconstruction accuracy. Compared to NMF and LDA, AA activates fewer components per document and exhibits larger cosine-angle separation, reinforcing its interpretation as a geometric topic modeling framework. These findings highlight the potential of AA as a

complementary alternative to probabilistic and centroid-based approaches for scientific text analysis. Future work may explore larger numbers of archetypes, increased document scale, and comparisons with neural topic models such as VAEs and Transformer-based architectures.

## 5 Conclusion

This study investigated the suitability of Archetypal Analysis (AA) as a topic modeling framework for high-dimensional categorical text data and compared its performance against established baseline methods, namely Non-Negative Matrix Factorization (NMF),  $k$ -means clustering, and Latent Dirichlet Allocation (LDA). With the exception of the custom AA implementation developed from scratch, all baseline models were implemented using built-in functions from the `scikit-learn` library to ensure reliable and standardized comparisons.

The experimental results demonstrate that Archetypal Analysis produces sparse, interpretable, and geometrically meaningful topic representations across both validation and conference datasets. In particular, AA consistently activates fewer components per document while maintaining competitive reconstruction accuracy, outperforming  $k$ -means and LDA in terms of interpretability and component separation. The results also confirm and extend the findings reported by Mørup and Hansen [3], validating the effectiveness of AA for text-based applications.

Non-Negative Matrix Factorization exhibited strong performance across both datasets and remained the most competitive baseline method. In contrast, LDA showed limited effectiveness under the experimental settings considered in this study, particularly in terms of sparsity and reconstruction quality. While LDA remains a well-established probabilistic topic model, its performance in this experimental context suggests that further tuning or alternative evaluation criteria may be required for meaningful comparison.

Overall, this work confirms that Archetypal Analysis is a viable and effective geometric topic modeling approach for scientific text analysis. The successful validation of the custom AA implementation and its competitive performance relative to established methods highlight its potential for further development. Future work may focus on improving the computational efficiency of AA, exploring larger numbers of archetypes, scaling to larger document collections, and extending the comparison to neural topic models such as variational autoencoders and Transformer-based architectures.

All experimental results and supplementary analyses are provided in the Appendix and the accompanying GitHub repository to ensure completeness and reproducibility. The AI Use Declaration and Ethics Statement are presented at the end of the paper in Section 7.4.

## 6 Acknowledgement

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## 7 Appendix

### 7.1 Code Chunks

#### 7.1.1 AA model

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**Algorithm 1** Alternating Optimization for Archetypal Analysis[2]

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**Require:** Data matrix  $\mathbf{X} \in \mathbb{R}^{N \times M}$ , number of archetypes  $K \in \mathbb{N}$

**Ensure:** Weight matrices  $\mathbf{S} \in \mathbb{R}^{N \times K}$  and  $\mathbf{C} \in \mathbb{R}^{K \times N}$  such that  $\mathbf{A} = \mathbf{C}\mathbf{X}$  and  $\mathbf{S}\mathbf{C}\mathbf{X} \approx \mathbf{X}$

```

1: Initialize archetypes  $\mathbf{A}$ 
2: while not converged do
3:   for  $n = 1, 2, \dots, N$  do
4:      $\mathbf{s}_n \leftarrow \arg \min_{\mathbf{s}_n} \|\mathbf{A}^\top \mathbf{s}_n - \mathbf{x}_n\|_2^2$ 
5:     subject to  $\mathbf{s}_n \geq 0, \|\mathbf{s}_n\|_1 = 1$ 
6:   end for
7:   Update archetypes:
                                      $\mathbf{A} \leftarrow (\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{S}^\top \mathbf{X}$ 

8:   for  $k = 1, 2, \dots, K$  do
9:      $\mathbf{c}_k \leftarrow \arg \min_{\mathbf{c}_k} \|\mathbf{X}^\top \mathbf{c}_k - \mathbf{a}_k\|_2^2$ 
10:    subject to  $\mathbf{c}_k \geq 0, \|\mathbf{c}_k\|_1 = 1$ 
11:  end for
12:  Update archetypes:
                                      $\mathbf{A} \leftarrow \mathbf{C}\mathbf{X}$ 

13: end while

```

---

```
import numpy as np
```

```
def project_to_simplex(v):
```

```

    """
    Project a vector v onto the probability simplex:
    {w | w >= 0, sum(w) = 1}.
    """
    v = np.asarray(v, dtype=float)
    n = v.shape[0]
    u = np.sort(v)[::-1]
    cssv = np.cumsum(u)
    rho = np.nonzero(u * np.arange(1, n + 1) > (cssv - 1))[0][-1]
    theta = (cssv[rho] - 1.0) / (rho + 1.0)
    w = np.maximum(v - theta, 0.0)
    return w

```

```
def solve_sls(B, y, max_iter=500, tol=1e-6):
```

```

    """
    Solve: min_s || B^T s - y ||^2 subject to s >= 0, sum(s) = 1
    using projected gradient descent (PGD).

```

```

    Note: B should be passed as the matrix (not transposed), and this
    function will compute B^T @ (B^T @ s - y) for the gradient.
    """

```

```

    K = B.shape[0] # B is (M, K) so we want K archetypes

```

```

s = np.ones(K) / K # start uniform

# For min ||B^T s - y||^2, gradient is: B @ (B^T @ s - y)
# Step size based on Lipschitz constant
step_size = 1.0 / (np.linalg.norm(B, 2) ** 2 + 1e-12)

prev_val = np.inf
for _ in range(max_iter):
    grad = B @ (B.T @ s - y) # Gradient for ||B^T s - y||^2
    s = s - step_size * grad
    s = project_to_simplex(s)

    val = 0.5 * np.linalg.norm(B.T @ s - y) ** 2
    if abs(prev_val - val) < tol * (1 + prev_val):
        break
    prev_val = val
return s

def archetypal_analysis(X, K, max_iter=50, tol=1e-5, random_state=0,
    verbose=True):
    """
    Archetypal Analysis (AA) via alternating optimization.

    Follows Algorithm 1:  $\min ||X - S^*C^*X||_{F^2} = \min ||X - S^*A||_{F^2}$ 
    where  $A = C^*X$  are the archetypes.

    Args:
        X: (N, M) data matrix (N samples, M features)
        K: number of archetypes
        max_iter: maximum iterations
        tol: convergence tolerance
        random_state: random seed

    Returns:
        S: (N, K) coefficients for data points in archetype space ( $s_n \geq 0$ ,  $\sum=1$ )
        C: (K, N) coefficients for archetypes in data space ( $c_k \geq 0$ ,  $\sum=1$ )
        A: (K, M) archetypes where  $A = C^*X$ 
    """
    rng = np.random.default_rng(random_state)
    N, M = X.shape

    # --- Initialize A with random rows of X ---
    idx = rng.choice(N, size=K, replace=False)
    A = X[idx, :].copy()

    S = np.ones((N, K)) / K
    C = np.ones((K, N)) / N
    prev_obj = np.inf

    for iteration in range(max_iter):
        # --- Step 1: Update S row by row ---
        # For each data point n, solve:  $\min ||A^T s_n - x_n||^2$  s.t.  $s_n \geq 0$ ,  $\sum(s_n) = 1$ 
        for n in range(N):
            S[n, :] = solve_sls(A, X[n, :])

```

```

# --- Step 2: Update A via least squares ---
#  $A = (S^T S)^{-1} S^T X$ 
StS = S.T @ S + 1e-10 * np.eye(K) # Add small regularization for
    stability
A = np.linalg.solve(StS, S.T @ X)

# --- Step 3: Update C row by row ---
# For each archetype k, solve:  $\min ||X^T c_k - a_k||^2$  s.t.  $c_k \geq 0$ ,  $\sum(c_k) = 1$ 
for k in range(K):
    C[k, :] = solve_sls(X, A[k, :])

# --- Step 4: Update A using C ---
#  $A = C * X$  (archetypes are convex combinations of data points)
A = C @ X

# --- Check convergence ---
#  $RSS = ||X - S * A||_F^2 = ||X - S * C * X||_F^2$ 
obj = 0.5 * np.linalg.norm(X - S @ A, "fro") ** 2

if verbose and iteration % 5 == 0:
    print(f"Iteration {iteration:3d}, RSS = {obj:.4e}")
if abs(prev_obj - obj) < tol * (1 + prev_obj):
    if verbose:
        print(f"Converged at iteration {iteration}, RSS = {obj:.4e}")
    break
prev_obj = obj

return S, C, A

```

### 7.1.2 NMF model

```

from sklearn.decomposition import NMF

def run_nmf(X, K, random_state=0, verbose=False):
    """Run Non-negative Matrix Factorization."""
    if verbose:
        print(f"\nRunning NMF with K={K}...")

    start_time = time.time()
    model = NMF(n_components=K, init='random', random_state=random_state,
                max_iter=200, verbose=0)
    W = model.fit_transform(X) # Document-topic matrix (N x K)
    H = model.components_     # Topic-word matrix (K x M)

    # Normalize W to get probability distribution
    W_norm = W / (W.sum(axis=1, keepdims=True) + 1e-10)

    elapsed = time.time() - start_time

    # Compute reconstruction error
    reconstruction = W @ H
    rss = np.linalg.norm(X - reconstruction, 'fro') ** 2

    if verbose:

```

```

        print(f" Completed in {elapsed:.2f}s, RSS = {rss:.4e}")

    return W_norm, H, rss, elapsed

```

### 7.1.3 K-means

```

from sklearn.cluster import KMeans

def run_kmeans(X, K, random_state=0, verbose=False):
    """Run K-means clustering."""
    if verbose:
        print(f"\nRunning K-means with K={K}...")

    start_time = time.time()
    model = KMeans(n_clusters=K, random_state=random_state, n_init=10,
                    max_iter=300)
    labels = model.fit_predict(X)
    centers = model.cluster_centers_ # (K x M)

    # Create one-hot encoding for hard assignments
    W = np.zeros((X.shape[0], K))
    W[np.arange(X.shape[0]), labels] = 1.0

    elapsed = time.time() - start_time

    # Compute reconstruction error
    reconstruction = W @ centers
    rss = np.linalg.norm(X - reconstruction, 'fro') ** 2

    # Compute clustering metrics
    silhouette = silhouette_score(X, labels)
    calinski = calinski_harabasz_score(X, labels)

    if verbose:
        print(f" Completed in {elapsed:.2f}s, RSS = {rss:.4e}")
        print(f" Silhouette Score: {silhouette:.4f}, Calinski-Harabasz: {
            calinski:.2f}")

    return W, centers, rss, elapsed, silhouette, calinski

```

### 7.1.4 LDA

```

from sklearn.decomposition import LatentDirichletAllocation

def run_lda(X, K, random_state=0, verbose=False):
    """Run Latent Dirichlet Allocation."""
    if verbose:
        print(f"\nRunning LDA with K={K}...")

    start_time = time.time()
    model = LatentDirichletAllocation(n_components=K, random_state=
                                     random_state,
                                     max_iter=20, learning_method='batch',
                                     verbose=0)

    W = model.fit_transform(X) # Document-topic distribution (N x K)
    H = model.components_     # Topic-word distribution (K x M)

```

```

elapsed = time.time() - start_time

# Compute reconstruction error (approximate for LDA)
reconstruction = W @ H
rss = np.linalg.norm(X - reconstruction, 'fro') ** 2

# Compute perplexity
perplexity = model.perplexity(X)

if verbose:
    print(f" Completed in {elapsed:.2f}s, RSS = {rss:.4e}")
    print(f" Perplexity: {perplexity:.2f}")

return W, H, rss, elapsed, perplexity

```

## 7.2 Method Explanations

### 7.2.1 Non-Negative Matrix Factorization (NMF)

Unsupervised learning algorithms such as Principal Component Analysis (PCA) and vector quantization can be viewed as matrix factorization methods with different constraints. Non-Negative Matrix Factorization (NMF), introduced by Lee and Seung [7], factorizes a non-negative data matrix into two lower-dimensional non-negative matrices:

$$\mathbf{X} \approx \mathbf{WH}.$$

Here,  $\mathbf{X} \in \mathbb{R}_+^{N \times M}$  denotes the document-term matrix, where each entry represents the frequency or TF-IDF weight of a word appearing in a document. The matrix  $\mathbf{W} \in \mathbb{R}_+^{N \times K}$  is the document-topic matrix, where each row indicates how strongly each of the  $K$  topics contributes to a given document. The matrix  $\mathbf{H} \in \mathbb{R}_+^{K \times M}$  is the topic-word matrix, in which each row represents a topic described by its most important words.

Because all values are non-negative, NMF provides an additive, parts-based representation that is intuitive for topic modeling. Through factorization, NMF attempts to uncover hidden structure in  $\mathbf{X}$  by expressing each document as a mixture of a small number of coherent topics [8].

The NMF objective function aims to find matrices  $\mathbf{W}$  and  $\mathbf{H}$  that minimize the reconstruction error measured by the Frobenius norm:

$$f(\mathbf{W}, \mathbf{H}) = \min_{\mathbf{W}, \mathbf{H} \geq 0} \|\mathbf{X} - \mathbf{WH}\|_F^2.$$

Lee and Seung [7] proposed multiplicative update rules that guarantee the objective function is non-increasing under each update. The updates for the Euclidean distance minimization are:

$$H_{a\mu} \leftarrow H_{a\mu} \frac{(\mathbf{W}^\top \mathbf{X})_{a\mu}}{(\mathbf{W}^\top \mathbf{WH})_{a\mu}},$$

$$W_{ia} \leftarrow W_{ia} \frac{(\mathbf{XH}^\top)_{ia}}{(\mathbf{WHH}^\top)_{ia}}.$$

The reconstruction error remains unchanged under these updates if and only if  $\mathbf{W}$  and  $\mathbf{H}$  are at a stationary point of the objective function.

It should be noted that, in this project, NMF is not implemented from scratch. Instead, the experiments employ the optimized NMF implementation provided by `scikit-learn`, as described in Section 7.1.2.

### 7.2.2 K-means

In text clustering, documents are represented as high-dimensional vectors using term frequency–inverse document frequency (TF–IDF) weighting. TF–IDF quantifies the importance of a word within a document relative to the entire corpus by combining two components: term frequency (TF) and inverse document frequency (IDF) [9].

For a given word  $w$  in document  $d$ , the TF–IDF weight is defined as

$$\text{TF-IDF}(w, d) = \text{TF}(w, d) \times \text{IDF}(w),$$

where

$$\text{TF}(w, d) = \frac{K}{T},$$

with  $K$  denoting the number of occurrences of word  $w$  in document  $d$ , and  $T$  the total number of words in document  $d$ . The inverse document frequency is given by

$$\text{IDF}(w) = \log \left( \frac{D}{\text{DF}(w)} \right),$$

where  $D$  is the total number of documents in the corpus and  $\text{DF}(w)$  is the number of documents containing the word  $w$ .

After vectorization, the  $k$ -means algorithm partitions the document set into  $K$  clusters by minimizing the within-cluster sum of squared distances,

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|_2^2,$$

where  $C_k$  denotes the  $k$ -th cluster and  $\mu_k$  is the centroid, computed as the mean of all document vectors assigned to that cluster. [4]

For text data, cosine similarity is commonly used instead of Euclidean distance in order to reduce sensitivity to document length and sparsity. Under this formulation, clustering is based on angular similarity between document vectors rather than their magnitude. [4]

In this study,  $k$ -means serves as a baseline clustering method. Due to its hard assignment mechanism, each document is associated with exactly one cluster, resulting in binary cluster memberships (0 or 1). While this limits interpretability and sparsity compared to Archetypal Analysis,  $k$ -means remains useful for identifying coarse-grained groupings of documents and dominant word patterns within clusters.

## 7.3 Results and Data

This section presents the experimental results, supporting data, and visualizations obtained during the study. It provides additional quantitative and qualitative insights that complement the methodological description given in the previous sections. The figures included in this section illustrate the behavior of the proposed models and highlight key patterns observed in the data.

All visualizations and plots were generated using the experimental code developed as part of this research and are intended to support the interpretation of the results.

### 7.3.1 Data

This research uses two types of datasets: a benchmark bag-of-words corpus and a collection of scientific conference abstracts. The benchmark dataset is used to validate and calibrate the Archetypal Analysis (AA) implementation developed from scratch, while the conference abstract dataset is used for the main experimental analysis.

**Bag-of-Words Dataset** The benchmark dataset is obtained from the UCI Machine Learning Repository<sup>2</sup>. In this study, the NIPS subset of the Bag-of-Words corpus is used. The dataset consists of  $N = 1500$  documents and a vocabulary of  $M = 12,419$  unique words, with approximately 6.4 million word occurrences in total. This dataset is commonly used in the literature and provides a reliable baseline for validating topic modeling methods.

**Conference Abstract Dataset** The second dataset consists of research paper abstracts collected from major machine learning conferences using the OpenReview API (<https://api.openreview.net>). A custom data collection script was developed to automatically download abstracts from conferences such as ICLR, NeurIPS, and ICML. This design allows flexibility in selecting different conferences and publication years for analysis.

The following commands illustrate how the abstract downloader can be used to retrieve different datasets:

```
# Smaller, manageable dataset (recommended for initial experiments)
python abstract_downloader.py --conference ICLR --year 2019

# Larger and more recent dataset
python abstract_downloader.py --conference ICLR --year 2023

# Cross-conference comparison for the same year
python abstract_downloader.py --conference ICLR --year 2019
python abstract_downloader.py --conference NeurIPS --year 2019
python abstract_downloader.py --conference ICML --year 2019
```

For the experiments reported in this work, the ICLR 2019 dataset was primarily used. Summary statistics for this dataset are shown below:

```
Conference: ICLR 2019
Total papers: 1419
Average abstract length: 1120 characters
Papers with keywords: 1315 (92.7%)
```

### 7.3.2 Results

In this study, Archetypal Analysis (AA) was evaluated under multiple experimental configurations in order to assess its geometric behavior and robustness across datasets. Experiments were conducted on two datasets: a Bag-of-Words (BoW) benchmark corpus and a collection of ICML 2019 conference abstracts. For both datasets, the number of documents was varied ( $\text{max\_docs} = 500$  and  $1000$ ), and the number of archetypes was set to  $K \in \{4, 5, 6, 10\}$ .

We first present Convex hull1 obtained from the Bag-of-Words dataset, which was used primarily for model validation. In this setting, clear convex hull structures are observed for  $K = 4$ ,  $K = 5$ , and  $K = 10$ , indicating that the learned archetypes correspond to extreme points of the data distribution. This behavior is consistent with the theoretical foundations of Archetypal Analysis.

We next consider the main experimental dataset consisting of ICML 2019 conference abstracts. The same experimental conditions were applied as in the validation experiments. As shown in Figure 3, the convex hull structure persists across different values of  $K$ , demonstrating that Archetypal Analysis consistently identifies extremal semantic patterns in real-world scientific text data.

Overall, these visualizations confirm that Archetypal Analysis successfully captures the geometric structure of both benchmark and real-world text datasets. The persistence of convex hull formations

---

<sup>2</sup><http://archive.ics.uci.edu/ml/datasets/Bag+of+Words>



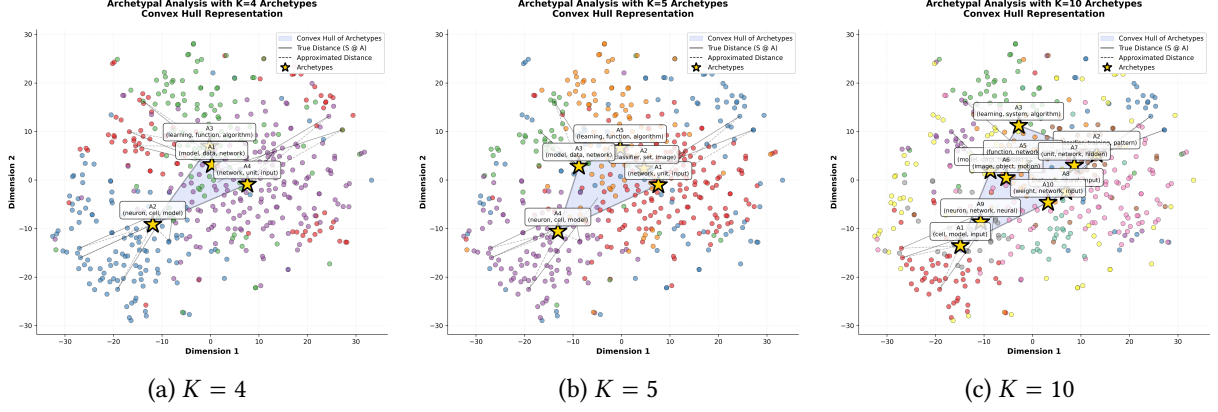


Figure 2: Two-dimensional visualization of Archetypal Analysis for the Bag-of-Words dataset with varying numbers of archetypes ( $K = 4, 5, 10$ ). The archetypes  $\{a_1, \dots, a_K\}$  define the vertices of the convex hull, while data points are represented as convex combinations of these extremes.

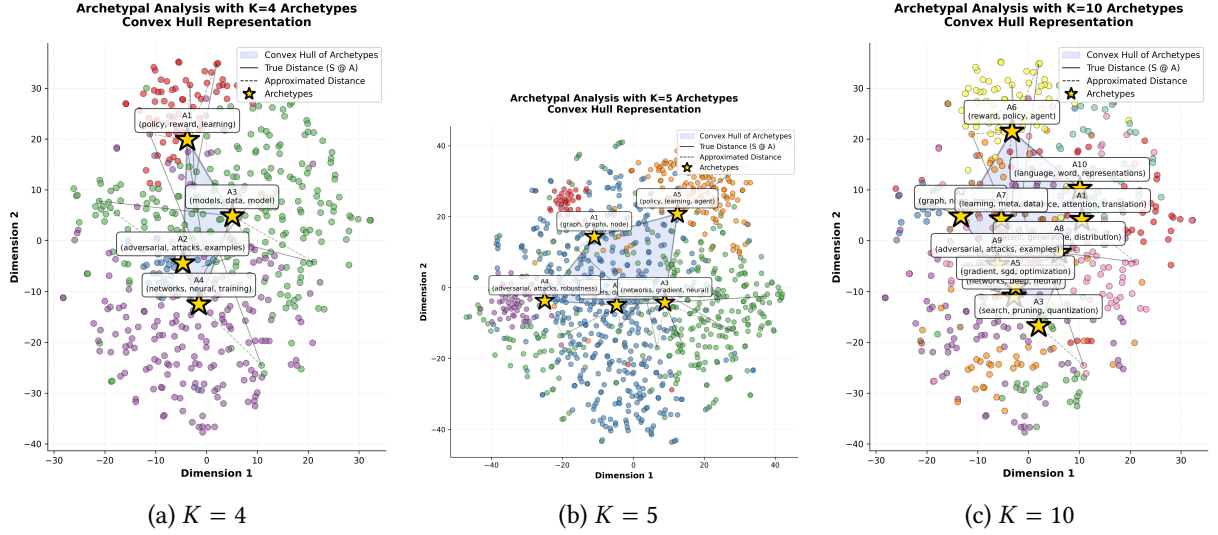


Figure 3: Two-dimensional visualization of Archetypal Analysis for ICML 2019 conference abstracts with varying numbers of archetypes ( $K = 4, 5, 10$ ). The presence of well-defined convex hulls indicates stable and interpretable archetypal structures in the learned representations.

across datasets and values of  $K$  provides empirical support for AA as a robust geometric topic modeling framework.

We now analyze the extracted word distributions for the case  $K = 10$ . Figure 4 presents the top words associated with each of the ten components obtained using Non-Negative Matrix Factorization (NMF), Archetypal Analysis (AA),  $k$ -means clustering, and Latent Dirichlet Allocation (LDA) on the Bag-of-Words dataset. The components are ordered by importance, and the grayscale intensity indicates the relative contribution of each word within a component.

We now turn to the experimental dataset consisting of ICML 2019 conference abstracts. Figure 5 shows the corresponding word distributions for the  $K = 10$  component models. The extracted components reveal clearly distinguishable thematic structures within the ICML corpus.

**Comparative Visualization Across Datasets** To provide a consolidated view of model behavior, Figure 6 summarizes the results obtained on the Bag-of-Words (NIPS-like) dataset, while Figure 7 presents the corresponding visualization for the ICML 2019 conference dataset. These visual summaries integrate both numerical metrics and structural patterns, enabling direct comparison across models and datasets.

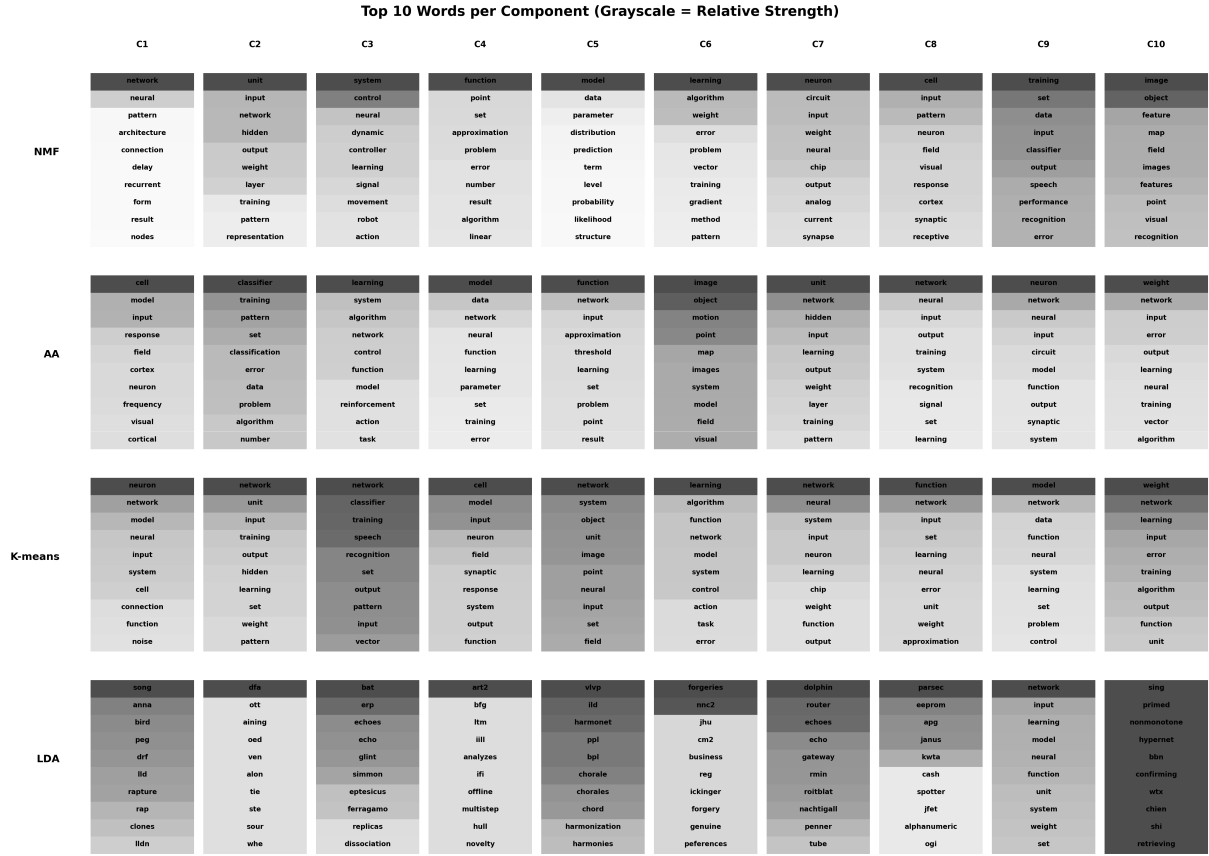


Figure 4: Top words for each component ( $K = 10$ ) extracted from the Bag-of-Words corpus using NMF, AA,  $k$ -means, and LDA. The average cosine angles between components are  $81.0^\circ$ ,  $64.9^\circ$ ,  $51.1^\circ$ , and  $18.9^\circ$ , respectively. These results indicate that AA extracts more geometrically distinct semantic directions than  $k$ -means and LDA. Furthermore, the proportion of non-zero entries in the document–component matrix  $S$  is 66.5% for NMF, 43.5% for AA, 10.0% for  $k$ -means, and 100% for LDA, highlighting the strong sparsity of AA relative to NMF and LDA.

A similar pattern is observed for the ICML 2019 dataset, as shown in Figure 7. Both AA and NMF exhibit moderate sparsity, with non-zero component usage of 46.6% and 51.7%, respectively. This behavior is clearly reflected in the heatmap structure, where semantic weight is concentrated in a limited number of components per document.

The results reported here correspond to the  $K = 10$  setting; however, analogous experiments were conducted for  $K = 4, 5$ , and 10 across both datasets. The qualitative trends observed in the visualizations remain consistent across these configurations. For completeness and reproducibility, the full set of experimental results and visual summaries is made available in the accompanying GitHub repository.

## 7.4 Use of Generative AI

During the course of this research project, generative artificial intelligence (AI) tools were used exclusively to support non-substantive aspects of the work. In particular, tools such as ChatGPT<sup>3</sup> were employed to refine grammar, improve sentence structure, and enhance the clarity and readability of the written text. In addition, Grok<sup>4</sup> was consulted to assist with code organization and the refinement of visualization layouts. All core algorithms, implementations, experimental designs, and analyses were developed independently by the author, with AI tools used only to improve presentation quality and

<sup>3</sup><https://chatgpt.com/>

<sup>4</sup><https://grok.com/>

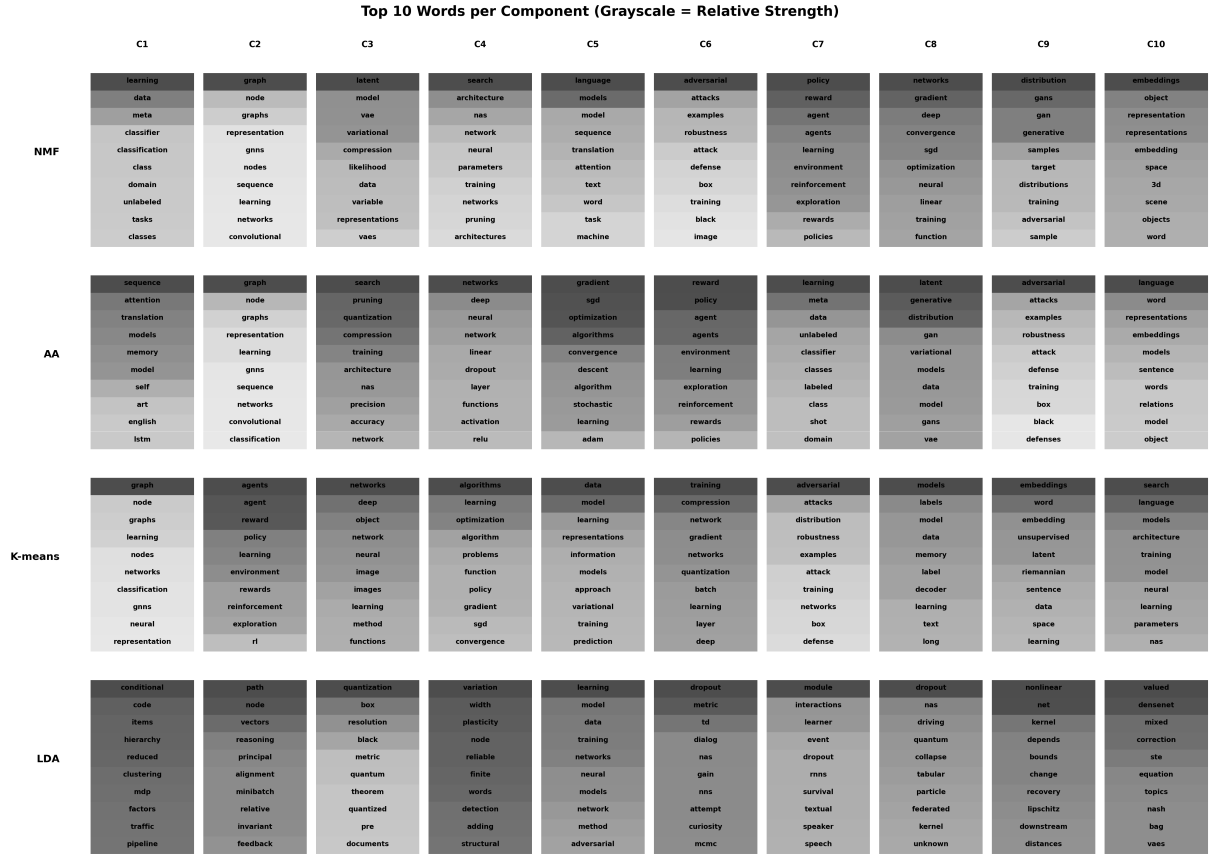


Figure 5: Top words for each component ( $K = 10$ ) extracted from the ICML 2019 conference abstracts using NMF, AA,  $k$ -means, and LDA. The average number of active components per document is 5.17 for NMF, 4.66 for AA, 1.00 for  $k$ -means, and 10.00 for LDA, indicating that AA achieves a favorable balance between sparsity and expressiveness. The average cosine angles are  $79.1^\circ$ ,  $73.8^\circ$ ,  $64.1^\circ$ , and  $46.4^\circ$ , respectively. AA also achieves the lowest Frobenius reconstruction error among the compared methods (21.20), demonstrating its ability to capture salient semantic structure while maintaining compact representations.

code cleanliness.

It is emphasized that all scientific contributions presented in this research project, including the research aims, methodology, mathematical formulations, experiments, results, and interpretations are entirely the author’s own work and were conducted independently under the guidance of the academic supervisor. No generative AI system was used to generate original research ideas, perform data analysis, derive results, or draw conclusions.

**AI Use Declaration and Ethics Statement** In accordance with academic integrity guidelines and institutional policies governing the use of artificial intelligence in scholarly work, the following declaration is made:

- **Use of AI Tools:** Generative AI tools (e.g., ChatGPT) were used solely for language refinement, structural improvements, and presentation-related support. No AI tool was used to generate original research content, experimental results, or scientific conclusions.
- **Independent Research Contribution:** All aspects of the research design, model development, experimental implementation, data analysis, and interpretation were carried out independently by the author.

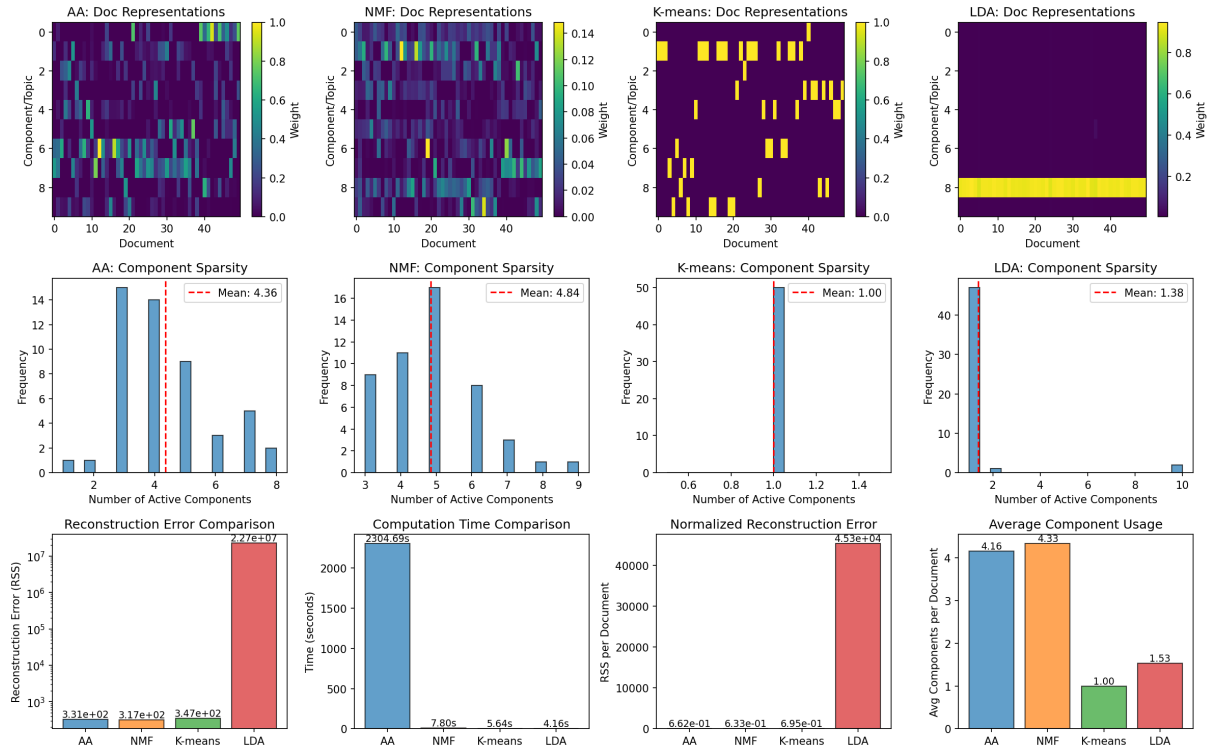


Figure 6: Summary visualization of topic modeling results on the Bag-of-Words dataset using Archetypal Analysis (AA), NMF,  $k$ -means, and LDA ( $K = 10$ ). The heatmaps indicate that AA and NMF activate components more broadly across documents compared to  $k$ -means and LDA. These visual patterns are consistent with the quantitative metrics, particularly sparsity, cosine-angle separation, and reconstruction error.

- **Ethical Compliance:** The use of AI tools in this project complies fully with the ethical standards and academic integrity guidelines of the university. The use of AI did not replace or misrepresent authorship or intellectual contribution.
- **Supervisor Oversight:** This research project was conducted under the supervision of the assigned academic supervisor, who provided guidance and feedback throughout the development of the work.

The author certifies that this Research Project represents an original academic contribution, with transparent and appropriate acknowledgment of the limited and ethical use of generative AI tools for linguistic and presentation-related enhancements.

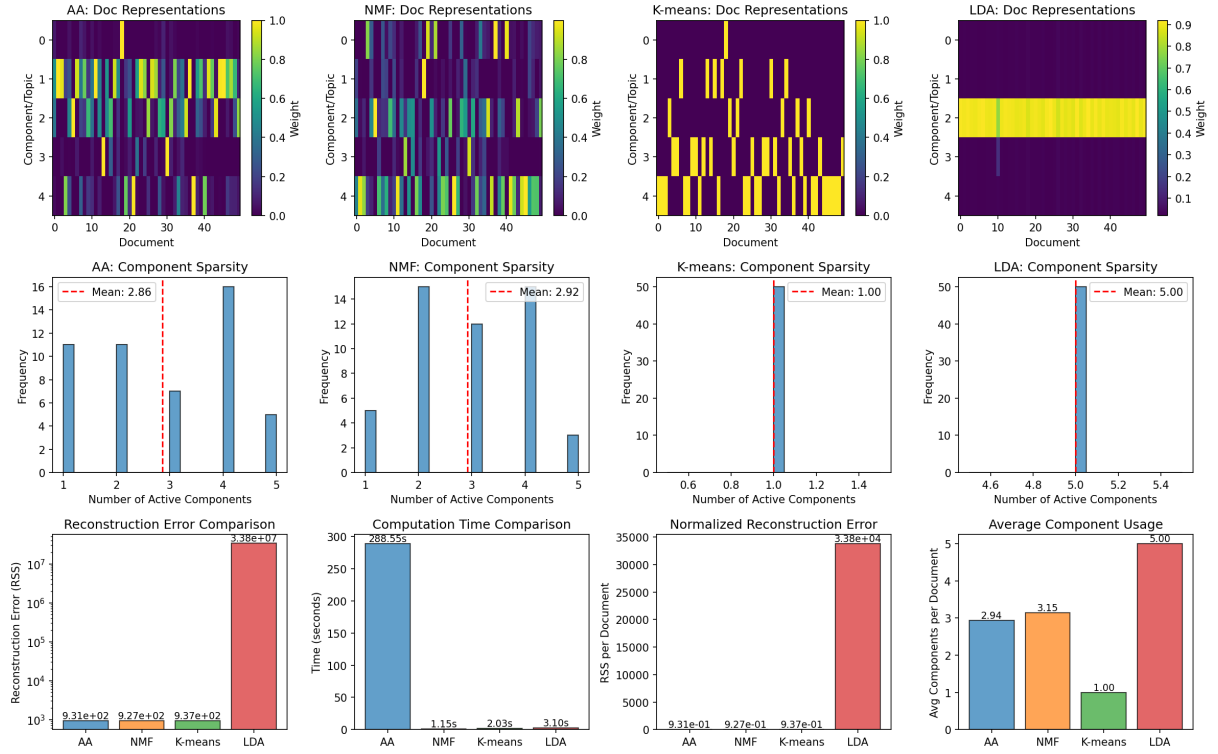


Figure 7: Summary visualization for the ICML 2019 conference dataset ( $K = 10$ ). Archetypal Analysis and NMF demonstrate comparatively strong sparsity and structured component activation, as reflected in the heatmaps. The reconstruction errors (RSS) are  $4.50 \times 10^2$ ,  $4.46 \times 10^2$ ,  $4.57 \times 10^2$ , and  $2.62 \times 10^6$  for AA, NMF,  $k$ -means, and LDA, respectively. While AA incurs a higher computational cost (524.52 s), it achieves a competitive balance between reconstruction accuracy, sparsity, and geometric interpretability.